Conclusion: a recovery-tested bridge with bounded claims

Conclusion: a recovery-tested bridge with bounded claims I

Active Fedference makes a deliberately narrow bridge between two bodies of work that are usually discussed with different objects and different standards of evidence. Active inference describes agents that infer and act within generative models; federated generalized Bayes describes how losses, divergences, and variational objectives can make decentralized inference less sensitive to bad information (Friston et al. 2017; Da Costa et al. 2020; Bissiri et al. 2016; Knoblauch et al. 2022). This paper does not dissolve those distinctions. It puts them into one categorical implementation, identifies the exact point at which they coincide, and then measures the consequences of moving away from that point.

The durable result is a recovery contract I

The central contribution is the recovery-tested contract. The KL/NLL client limits of the declared generalized-Bayes construction recover the closed-form Bayes update, while the zero-robustness server branch recovers the project log-linear pool, with maximum measured deviations of 5.55e-17 and 0 (eq. 6; eq. 7). This is stronger than a verbal analogy because the limited identities are stated in the formalism, implemented in the core, and checked by executable invariants. Under the explicit shared-support, posterior-log-potential, and fixed-weight bridge, the server pool specializes Eq. 7's message-combination term; it is not an assertion that the active-inference and robust-Bayes literatures share all assumptions, objectives, deployment meanings, or the complete source protocol.

The durable result is a recovery contract I

That distinction matters historically and methodologically. Logarithmic pooling has a substantial literature as an aggregation rule for expert distributions, including its connections to external Bayesianity, product-of-experts constructions, and KL-based opinion pooling (Genest and Zidek 1986; Genest et al. 1986; Hinton 2002; Abbas 2009; Dietrich 2021). FedGVI contributes a generalized-Bayes perspective in which the loss and divergence determine what robustness means (Mildner et al. 2025). The contribution here is therefore a scoped recasting: the project's categorical log-linear-pool specialization is the non-robust corner of the implemented family under its stated bridge assumptions, and the corner becomes a testable boundary condition for future robust extensions. This does not identify the complete source belief-sharing protocol with FedGVI.

What the evidence establishes away from the corner I

The standard-Bayes studies provide a necessary baseline rather than ornamental background. Communication changes mean free energy by 3.3109 nats, Dirichlet learning reduces KL from 3.4231 to 0.0027, and Bayesian model reduction selects the declared redundant-pruning candidate. These results recover the declared mechanisms that make belief sharing scientifically interesting while keeping the source relationship bounded to the stated categorical protocol (Friston et al. 2024). The extension studies then show that communication is not automatically beneficial in every information geometry: disjoint views can make sharing valuable, whereas a complementary moving-world control can make additional pooling unnecessary or mildly costly.

What the evidence establishes away from the corner I

The contamination study adds a second lesson. The server-side heuristic is regime-dependent: robust operating points can give up a little efficiency when contamination is weak and recover that cost when the declared attack is severe. At the most severe swept rate, the highest pooled robust mean reaches 0.9880 against the standard pool's 0.6928 (tbl. 3); at the verdict rate, the matched, BH-adjusted comparison of sec. 7.5 gives 0.9867 against 0.9021. The result is therefore an operating-point contrast, not a ranking that holds for every contamination rate, attack target, hidden state, or calibration regime. This interpretation follows the simulation-study principle that the data-generating mechanism, estimand, and Monte Carlo unit should be declared before a numerical result is treated as general evidence (Morris et al. 2019).

What the evidence establishes away from the corner I

The three robustness axes give the result its proper meaning. The client-side generalized-Bayes update is the FedGVI-faithful, theorem-bearing axis under its matching loss and regularity assumptions. The server-side `robust_aggregate` is the sharp empirical heuristic; its proven property here is the recovery limit, not a transferred client-side bounded-influence theorem. The `variational_aggregate` is the conservative complement: it descends a stated aggregation objective and has a proven raw effective-weight bound, but its objective-backed control does not make it the peak-accuracy display leader. Separating these axes prevents a familiar failure in interdisciplinary work: a guarantee proved for one operator is silently attached to another operator because both are described with the same word, “robust.”

Why the bridge matters for active inference I

Belief sharing is not merely a communication convenience. In an active-inference system, a shared posterior can change expected free-energy calculations, model comparison, and subsequent action selection. A robustification at the fusion step can therefore alter behavior even when each local generative model is unchanged. Conversely, a server rule that suppresses an anomalous belief may also suppress a rare but correct observation. The relevant scientific question is not simply whether a robust curve rises; it is which assumptions about competence, independence, support, and action-relevant uncertainty the fusion rule encodes (Genest and Zidek 1986; Tresp 2000; Heins et al. 2024).

Why the bridge matters for active inference I

This is why the recovery corner is a useful organizing device. It gives a common reference behavior before robustness is introduced, makes the cost of a server intervention measurable, and lets later work compare new aggregation rules against an interpretable baseline. The bridge also keeps the distinction between inference and infrastructure visible. The current federation tests show that serialized beliefs can travel through the declared local transport and return a bit-identical consensus, but they do not turn a mathematical aggregation identity into a claim about secure, fault-tolerant, or privacy- preserving deployment (McMahan et al. 2017; Blanchard et al. 2017; Pillutla et al. 2022).

What remains unproved I

The evidence does not establish universal Byzantine tolerance, truth recovery, calibration, or an optimal robustness parameter. The primary intervals are conditional on the fixed hidden state and attack target, and the nested trial and seed structure is reduced at the declared unit rather than treated as a larger independent sample (Koehler et al. 2009; Loy and Korobova 2021). The categorical state space is the object of the proof and experiment; continuous or hybrid state spaces are a separate mathematical extension. The neural classification complement uses point-estimate weights, so it does not establish full posterior FedGVI behavior at the scale of the source experiments.

What remains unproved I

These are not defects to be hidden by a more expansive title. They define the proper contribution: an executable categorical bridge, a recovery certificate, an explicit map of theorem-bearing and heuristic components, and conditional evidence about contamination behavior. Robust-statistics language such as influence and breakdown remains useful for describing the failure modes, but a finite simulation sweep is not by itself a general estimator-level robustness theorem (Huber and Ronchetti 2009). The same discipline applies to historical and conceptual scholarship: early work on probability, inverse inference, utility, and collective judgment supplies lineage, not evidence for modern KL, FedGVI, or adversarial-federation claims.

A falsifiable research program I

The next stage should be organized around boundary conditions rather than a larger collection of demonstrations. First, derive a server objective whose minimizer is competitive with `robust_aggregate` while retaining the variational rule's effective-weight control. Recent work on closed-form generalized variational objectives, logarithmic-pool weighting, and robust divergence-weighted federation provides relevant mathematical constraints (Nguyen and Westerhout 2026; Carvalho et al. 2023; Li et al. 2022). A useful candidate must recover the standard log-linear pool at zero robustness, state which quantity is bounded, and fail visibly when those assumptions are violated.

A falsifiable research program I

Second, broaden the primary estimand across attack targets, hidden states, adaptive adversaries, calibration conditions, and model classes. The decisive falsifier is not a lower average score on one new grid; it is failure of the claimed robustness advantage, recovery identity, or stated uncertainty calibration under a pre-registered extension of the data-generating mechanism. Third, promote the point-estimate neural complement to a posterior-parameterized FedGVI study at source-comparable scale, and test whether the client-side bounded-loss behavior survives capacity, optimization, and posterior-family changes. Finally, move from local transport to multi-machine execution with explicit threat models, authentication, failure handling, and privacy claims; none should be inferred from the current bit-identity result.

Final position I

The strongest conclusion is consequently neither that robust belief sharing has been solved nor that the bridge is merely metaphorical. In the declared categorical setting, the standard-Bayes client limits and project log-linear-pool server identity are tested recovery conditions; away from them, server behavior is measurable, regime-dependent, and partitioned into theorem-bearing, objective-backed, and heuristic claims. That is a modest result, but it is a useful one: it supplies a reproducible starting point from which stronger theorems, broader state spaces, independent implementations, and real federated deployments can be judged without losing the baseline they are meant to extend.